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Brain Sizes, Surfaces, and Neuronal Sizes of the Cortex Cerebri: A Stereological Investigation of Man and His Variability and a Comparison With Some Mammals (Primates, Whales, Marsupials, Insectivores, and One Elephant)

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ABSTRACT **This study deals with the stereological estimation of macroscopic sizes of brain and cortex, i.e., volume, surface, and folding, and of microscopic neuronal sizes, i.e., density, mean size, size distribution, and number of neurons. The results show that the degree of variability in man amounts to about 15%. A decrease in volume of the different gray structures can be observed in man after the age of 65 years. The surface, folding index, and length of convolution do not alter with aging. The comparison with mammals of various sizes allows the conclusion that there is a high correlation to brain size for nearly all macroscopic values. Man and elephant, however, have a cortical surface which is, in comparison with whales, relatively small. In contrast, whales have very small cortices compared with man.**

At the cytoarchitectonic level, the neuronal density has a correlation to brain size. Contrary to other mammals, the primates and man have a high fraction of small granular neurons, especially in layer 4. The assumption that the number of cortical neurons beneath a given surface area of cortex is the same in all mammals cannot be verified, especially in those with large brains. The allometric connection between brain size and parameters is not valid for all measurements (e.g., thickness of cortex, mean size of neurons, perikaryal size distribution, and glial density). Yet some other measurements are well correlated.

INTRODUCTION

For about the last 30 years, stereological procedures have been increasingly used for investigations of the nervous system. Stereology enables us to estimate size in all dimensions from macroscopy to light to electron microscopy. As to the macroscopic level, Elias et al. (1967b) introduced the measurement of the surface area of the convoluted cerebral cortex. The estimation of volume by point counting has been performed since 1956 by Haug (1956) and Hennig (1958). In the beginning, Haug used this procedure for the perikaryon volume-fraction or gray-cell coefficient of the cortex cerebri (Economo, 1926).

Since 1970, point counting has also been performed for the volume estimation of the different gray structures in the total brain when working macroscopically (Haug, 1970a,b; Decker, 1979). In order to measure the mean neuronal density in certain parts of the brain, the counting of nerve cells was also stereologically facilitated by Haug (1967b). Further important stereological estimations of the brain were reported upon by Stephan (1960), Schlenska (1969), and Kühl et al. (1982).

The results of these and other similar investigations were published in diverse places, and it has been necessary to combine the main efforts of these investigations. The size of brain and its parts were mainly published regarding man (Elias et al., 1967a,b; Haug, 1970a; Schlenska, 1969; Eggers et al., 1984a) and some mammals (Elias et al., 1967a, 1969; Elias and Schwartz, 1969, 1971; Haug, 1970a, Wiedemeyer, 1974), but some of the results of Elias have not been published. At the cytoarchitectonic level we find morphometric results for man (Haug,1953; Haug et al., 1984; Barmwater, 1982; Kühl, 1981; Nass, 1985; Sass,1982a), for large mammals (Haug, 1967a; Jacobs et al., 1984; Gunschmann, 1972; Tjan, 1974), for small mammals (Moral and Haug, 1982; Stephan and Andy, 1969; Stephan et al., 1984; Wiedemeyer, 1974), and for the comparison of mammals (Haug, 1956, 1958, 1960, 1976, 1981b; Stephan, 1972).

The findings of our investigation must be combined in order to interpret general aspects. The ensuing results should be compared with other findings and with various assumptions about the connections between the structural sizes. The mathematical theory, according to which different sizes of brain and body are connected, in animals, was finally formulated by Dubois (1914). The sizes are connected over many decades of exponential powers by the formula of "allometry" according to the exponential regression $y = b \cdot x^a$. Many papers have been presented on this subject. Jerison (1974) has given a survey on this theory.

A second kind of comparison, concerning, in particular, the information on the extent of evolutionary development, was the introduction of indices in which different sizes were connected by ratios. The goal of such connections is to find an index which expresses a progressive evolution of a certain neuronal system. The first steps were made by Wirz (1950). In recent decades

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TABLE 1. Number of human brains in our investigation

	Macroscopy				Microscopy area				
Author	Male	Female	Surface	Volume	06	07	11	17	20
Schlenska	4	3	7	7					
Elias	10	10	20	—					
Eggers	6	6	12	12					
Holoubek[1]	10	—	10	10					
Haug	75	43	—	—	59	51	60	50	20

[1]Preliminary results.

Stephan (1972, 1976), Stephan and Andy (1969), Bauchot (1978), Jerison (1974), Pirlot (1974), and Hofman (1984) have advanced this project. At the same time first efforts were made in microscopic allometry of neuronal structures. The density of neurons showed a connection with brain size (Tower, 1954; Bok, 1959; Haug, 1958, 1960, 1976). Further values of the neurons such as the gray-cell coefficient (Economo, 1926; Haug, 1958) and neuronal number below the surface were examined in order to get a useful answer regarding evolutionary steps (Bok and Taalman Kip, 1939; Rockel et al., 1980).

Besides primates including man, smaller mammals as well as animals with very large brains like dolphins and the elephant must be taken into consideration, and the results must be compared with the above investigations and with the information on brain sizes compiled by Blinkov and Glezer (1968), Jacobs et al. (1984), Jansen (1973), Kesarev et al. (1977), Langworthy (1932), Morgane and Jacobs (1972), and Morgane et al. (1980). These results may help to interpret our own data.

The aim of this examination is to survey our present knowledge of cortical structure, which has been reached by the introduction of stereological procedures. The results reported may also possibly give an important insight into the aging of the human brain cortex. However, data at the level of electron microscopy will be mentioned only briefly.

MATERIAL AND METHODS

The material of this investigation was collected by the groups of Elias and Haug. Not all the results have been published. Details as to the material can be found in the relevant literature. The human material is described by Elias and Schwartz (1971), Haug (1967a, 1970a, 1984), Schlenska (1969), and Eggers et al. (1984a). Table 1 contains the human material which will be reported. Table 2 refers to the animal material.

The following authors used the collection of Haug: Barmwater (1982), Fischer (1982), Gunschmann (1972), Kühl (1981), Nass (1985), and Sass (1982a). The number of areas investigated was distinctly higher than the total number of brains. This is due to those brains which have been evaluated in more than one area.

The animals for the investigations were also collected by Elias and Haug. Most of the whales (Odontoceti of different sizes) and marsupials were sampled by Elias. The primates and insectivores were collected in Europe. The specimens listed in Table 3 are included in the results.

The material collected was fixed at different places in the world. Therefore the shrinkage of the single specimen is not reliably determined. According to our experience, we must expect that the sizes estimated in a specimen may deviate from the actual size of the species investigated. The differences, however, are surely small, as shown by the estimation of the variance in man. The variability in man is about 15–20% above and below the mean values (Haug et al., 1984; Eggers et al., 1984a). The differences between the animals of various families are in the magnitude of a power of four. Therefore, we feel that a comparison is allowable. This point will be discussed after the report of the results.

The procedures used are described in detail in the stereological literature (Elias, 1962, 1967; Haug, 1956, 1967b, 1979, 1981a, 1982; Haug et al., 1971; Weibel, 1979). The following methods are the basis of our investigation:

1) The point counting of the volume and of the volume fraction according to the DELESSE principle was used for the estimation of the volumes of the brain parts in the macroscopic and microscopic dimensions.

This procedure used in macroscopy was described by Haug (1970a) and many other examiners. The total volume corresponds with the total number of grid points found in the cuts through the brain structures. The volumes of the single brain parts were calculated by the relevant point-fraction. The point counting was also performed by the estimation of the perikaryon volume fraction (Haug, 1956). For about the last 10 years, however, the investigations have been made mainly indirectly through size distribution (see 6 below).

2) The measurement of the surface was performed by point counting with the formula

$$S_V = 2 \times I_A \quad (1)$$

For the measurement of the cortical surface we used the deduced formula which was introduced by Elias (1967):

$$S = 2 \times I \times T \times D \quad (2)$$

In this formula T means the thickness of the slides or sections estimated and D the distance of the lines on the grid thrown on the slide or its photograph. This kind of stereological procedure is sensitive to the orientation of the structures in the slides. Therefore the grid used was turned several times.

3) The counting of neurons and glial cells was performed according to the formula introduced by Floderus (1944).

The counting itself is restricted to the nucleoli of the nerve cells. Furthermore, we use a thickness of section 20 μm. The thickness of each section is measured by

TABLE 2. The various species of this investigation including the brain volume (Br-vo) in cm^3 and our register number[1]

	Abbr.	Latin name	Common name	Br-vo	Reg. no.
1	DD	*Delphinus delphinus*	Common dolphin, mean	715	275
2	DT	*Tursiops truncatus*	Bottlenosed dolphin, mean	1,345	272
3	DG	*Globicephala macrorryncha*	Pilot whale, mean	2,720	270
4	DP	*Phocaena phocaena*	Common porpoise	483	271
5	DC	*Pseudorca crassidens*	False killer whale	3,650	274
6	EL	*Loxodonta africana*	Elephant	4,148	260
7	UE	*Equus caballus*	Horse	510	251
8	US	*Ovis aries*	Sheep	140	252
9	CL	*Panthera leo*	Lion, mean	259	150
10	CC	*Canis familiaris*	Dog	63.7	205
11	CC	*Canis latrans*	Coyote	85	205
12	CF	*Felis domestica*	Cat	28.7	200
13	CR	*Procyon lator*	Raccoon	43	154
14	CV	*Vulpes vulpes*	Fox	45	152
15	H	*Homo sapiens*	Human, mean	1,300	000
16	OS	*Pan troglodytes*	Chimpanzee	433	102
17	OG	*Gorilla gorilla*	Gorilla	429	102
18	OP	*Papio duguera*	Baboon	184	101
19	OR	*Macaca mulatta*	Rhesus monkey, mean	88	101
20	OC	*Cercopithecus aetiops*	Vervet monkey	64	106
21	OT	*Cercopithecus talapoin*	Talapoin, mean	36	106
22	NA	*Ateles ater*	Spider monkey	106	107
23	NL	*Lagothrix lagotricha*	Wooly monkey	101	107
24	NC	*Cebus albifrons*	Capuchine monkey	57	107
25	NS	*Saimiri sciurius*	Squirrel monkey, mean	22.4	108
26	NT	*Aotes trivirigatus*	Night monkey	18.9	107
27	NJ	*Callithrix jacchus*	Marmoset, mean	9.4	109
28	PL	*Lemur div.*	Lemur, mean	25.8	111
29	PR	*Lemur ruficaudatus*	Weasel lemur	7.4	112
30	PM	*Microcebus murinus*	Mouse lemur	1.9	113
31	TG	*Tupaia glis*	Tree shrew	3.0	114
32	RC	*Oryctolagus cuniculus*	Rabbit	11.2	230
33	RR	*Rattus norvegicus*	Rat	1.7	221
34	RM	*Mus musculus*	Mouse	0.3	220
35	RG	*Cavia porcellus*	Guinea pig, mean	3.9	231
36	IE	*Erinaceus europaeus*	Hedgehog, mean	3.7	180
37	IT	*Talpa europaeus*	Mole	1.4	181
38	IS	*Sorex*	Shrew, mean	0.24	183
39	MK	*Wallabia*	Kangaroo, mean	42.5	170
40	MO	*Didelphis virginiana*	Opossum, mean	7.8	173
41	MM	*Marmosa mitis*	Mouse opossum	1.2	172
42	ME	*Tachyglossus*	Echidna	11.0	171

[1]In all primates including prosimians the lateral convexity of the frontal cortex was microscopically estimated. For further details see Table 3.

TABLE 3. Mammalian brains used for macroscopy and microscopy of the cortex cerebri[1]

		Macroscopy			Microscopy area			
Animal	Abb.	n	Surface	Volume	Fro.	Par.	Occ.	Tem.
Loxodonta africana	EL	1	1	1	1	1	1	1
Pseudorca crassidens	DC	1	1	1	1	—	1	—
Globicephala maelas	DG	3	3	2	2	1	2	1
Tursiops truncatus	DT	5	5	3	1	1	1	1
Delphinus delphinus	DD	2	2	2	2	2	1	1
Phocaena phocaena	DP	1	1	1	1	1	1	1
Didelphis virginiana	MO	1	1	1	—	1	—	—
Erinaceus europaeus	IE	1	1	1	—	1	—	—
Talpa europaeus	IT	1	1	1	—	1	—	—

[1]Abbreviations: Fro., frontal lobe; Par., parietal lobe; Occ., occipital lobe; Tem., temporal lobe.

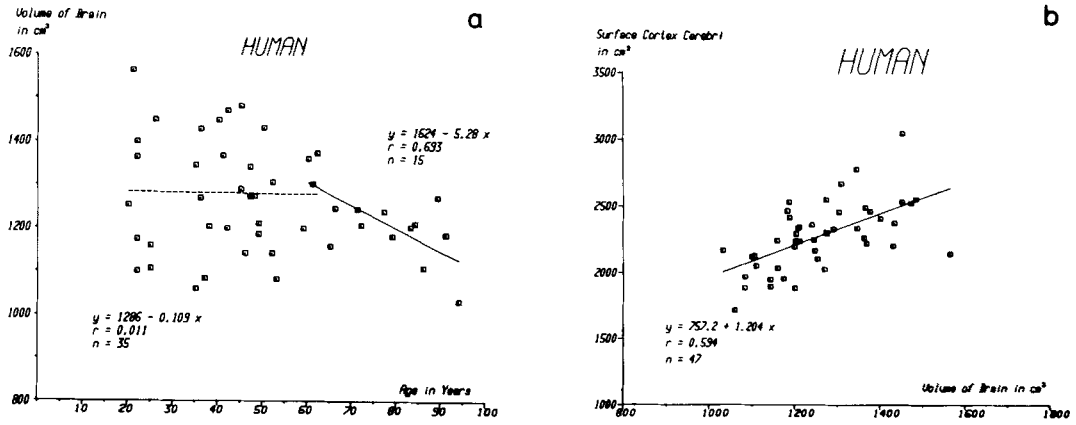


Fig. 1. a: The linear regression between the human brain size (ordinate in cm^3) and age (abcissa in years). The correlation between 20 and 65 years is insignificant (values of Haug and co-workers). The correlation after the age of 60 is significant ($P = .01$). b: The surface area of the cortex cerebri compared with brain size in humans. The linear regression is highly significant ($P = .001$) in spite of the variance (values of Elias and co-workers and Haug and co-workers).

high-power oil-immersion microscopy. In this way the remaining bias in estimation is minimized.

4) Within the last 10 years the investigations of the neurons have been performed in our group with the help of digitizer instruments (MOP and a Videoplan, Kontron).

We used these instruments in the macroscopic dimension and in microscopy by measuring over a drawing-mirror and by tracing the figure of the cells with an illuminated tracer (Haug, 1981a). The perikarya of those cells with a nucleolus are traced by following their largest visible circumference. This procedure has the advantage that there is a high possibility to measure the largest actual circumference. In this way we obtain an unfolded size distribution (sd). The measurements are restricted to those cells with a nucleolus inside the grid of observation. In fact we also observe the Gundersen's rule (1978) for the edges.

5) The surface and volume estimations with digitizing instruments are performed for the surface (S) by the boundary length (B) and the thickness of slice (T) according to the formula

$$S = \frac{4}{\pi} \cdot B \cdot T \qquad 3)$$

The volumes are calculated by the areal fractions found in the total structure image and the single substructures (Weibel, 1979).

6) The sd was evaluated from the projection area of the perikarya.

With the computer it is possible to choose a classification which can be adapted to the actual size. In general we use a classification with 12 classes. We measure at least 1,000 cells per specimen. In order to adapt the differently arranged size classifications, we calculated with a total of 26 classes for the comparison in mammals (Haug, 1979).

We generally use a classification of neuron projection area in groups (classes) of 20 μm^3. In the classes numbered higher than 12, the class width was enlarged in such a way that with rising numbers the width increased. As a consequence, the numbers in the larger classes rose according to the width. For graphics it was thus necessary to reduce the number of cells in these classes with respect to the class width.

7) During the microscopic estimation, the fields of measurement are arranged perpendicularly to the surface of cortex and are called rows (Haug et al., 1971).

Each field lies in apposition to the preceding field. Furthermore, the distance of the fields from the surface is registered as numbered lines. The first field is apposed to the surface of the cortex. In this way we can get an insight into the cortical layers. Later on we introduced coordinates for the so-called "gravity points" of the measured perikaryon projection areas. We can sample coordinates of cells over a field of a maximum size of 3.5–6 mm^2. With these coordinates we are able to print scatter-diagrams of the distribution of cells within a satisfactory extension of an area for all cells, or for neurons selected according to a certain type or a certain size (Haug, 1982).

8) The data collection performed for each neuron made it possible to calculate the real size distribution for the perikaryon projection area and for the volume of each cell.

With the use of the formula of Kühl et al. (1982), it is possible to calculate the perikaryon volume of each perikaryon. The sum of all neuronal volumes in a space yields the perikaryon-volume fraction (vo%). The estimation of this value by point counting is also possible and amounts to nearly the same quantity. The perikaryon-volume fraction is the reciprocal value of the "Grauzellkoeffizient" (gray-cell coefficient) introduced by Economo (1926).

9) The shrinkage during embedding was estimated; and it was discovered that by using methylbenzoate, an age-dependent shrinkage of the brain tissue occurred (Sass, 1982b; Eggers et al., 1984b).

The term used for the shrinkage is called residual volume, which describes the size as a fraction or as a percentage of the fresh volume. During our calculation

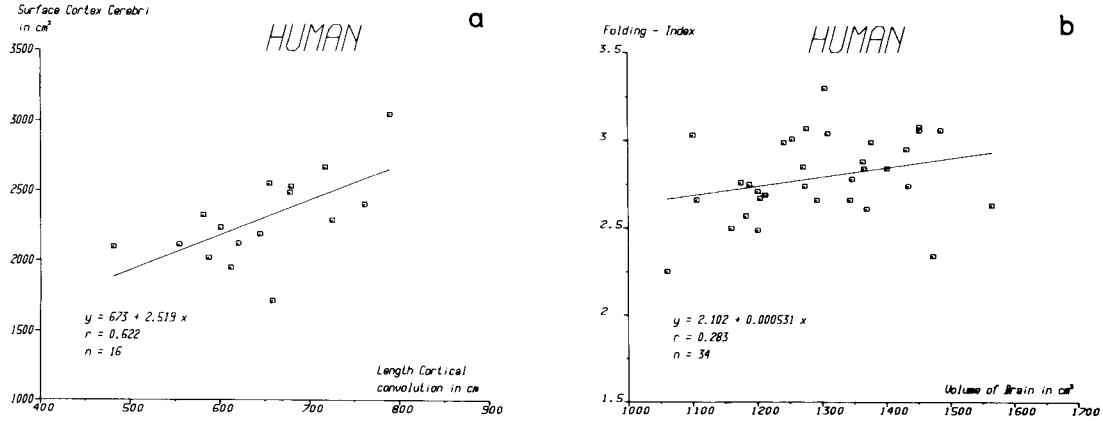


Fig. 2. **a:** The relation of the surface of the total cortex to the length of gyral convolution in humans. The correlation is significant ($P = .01$) (values of Elias with co-workers). **b:** The relation between the volume of brain and the folding index. The linear regression in the human is less significant ($P = .1$) (values of Elias and co-workers and Haug and co-workers).

of results, we perform a correction of the data. The results mentioned in this paper are calculated with regard to the so-called fresh volume. For brains in which the shrinkage could not be measured, we calculate with a factor corresponding to a residual volume of 0.60.

10) The values of the macroscopic and microscopic evaluation can be combined in order to obtain information as to sizes which cannot be measured directly.

Two sizes are important in our work. One is the average thickness of the cortex in a specimen. It is calculated by dividing the volume by the surface of cortex cerebri. The other size is the total amount of nerve cells in the cortex cerebri, and it can be computed by a multiplication of the neuronal density by the cortical volume. Such values were calculated for the single specimen. In this way we gain an insight into the variability of the individual number of cortical neurons. The size distribution and the number of differently sized neurons can also be calculated.

11) All our values were statistically secured by the usual statistical methods of the t-values or regression.

In the comparisons, we used the linear, logarithmic, and exponential regression function, depending on the range of values and the best approximation.

RESULTS

The results are presented in the following manner: Each part begins with the results in humans. In this way information is gained on interindividual variation. The assumption is made that the variability within each of the other species is about the same as in humans. The human values for the comparative part are represented by the average of our own results. In the comparison of the mammalian results it must be taken into account that the data from one specimen per species is not the mean of the species but may lie within a certain, yet unknown, variability. Therefore, the examiner is unable to estimate the actual position of the specimen investigated within the field of variance. However, it can be assumed that the results, as those of man, are nearly free from the uncertainties of variance. The comparison of the different animals and men will be made mainly with the regression functions.

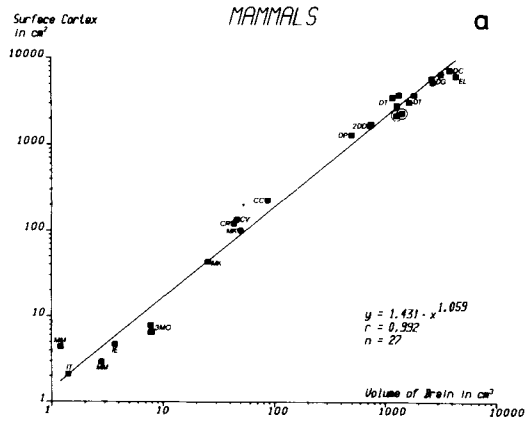
The Macroscopic Sizes of the Brain—Variance and Comparison

In this section the volume of the total brain (Br-vo) and of some parts are presented as well as the size and configuration of the surface of cortex (Cc-sa; surface area in cm^3), the folding index (Cc-fi), and the length of convolution (Cc-le) of the cortex. Both the latter expressions were introduced by Elias et al. (1967b). The Cc-fi is an expression for the folding of the brain and means the ratio between the total surface including all deep parts of the convolution and the external surface of the cerebral cortex. Cc-le represents the length of those convolutions which can be seen on the exterior surface of the cortex.

The area of the surface of the cortex was estimated for the first time by Wagner (1864). His results are very similar to the new measurements. The complicated estimations were rarely used. The introduction of the new stereological procedures, however, opens a new epoch of measurement of surface structures.

The variability of human brain size has been described by Haug (1984) in detail. The brain size is influenced by the increasing body size resulting from the secular acceleration. Since about the end of the last century, this increase in size has been about 1 mm per year. A secular acceleration was not observed before the middle of the 18th century. The size of body had been nearly constant for about 2,000 years. The present secular acceleration of the body size of 1 mm per year also has led to an increase of the brain size of about 0.6 cm^3 per year, or to 50–60 gm per century (Haug, 1984).

Figure 1a shows the variation of human brain size in relation to age. We do not differentiate between males and females in the graph. The brain size of the adult man is constant up to the age of 60 to 65 years. Only a slight decrease, not significant, can be observed between the ages of 20 and 65 years. Later, a distinct decrease of size can be observed: it is about 6 to 8% less at the age



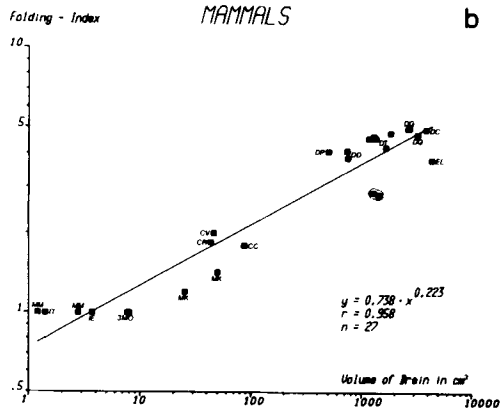


Fig. 3. a: There is an exponential regression between the surface cortex (abscissa) and the brain size (ordinate) in mammals. The human values are circled. For abbreviations, see Table 2. The correlation is significant (P = .001) (values of Elias and co-workers and Haug and co-workers). b: The folding index in mammals has a logarithmic regression and is highly significant (P = .001). It should be noted that the values for man and elephant deviate from the mean correlation line in the lower direction (values of Elias and co-workers and Haug and co-workers).

of 90 years. This size decrease is significant (P = .01 in spite of the relatively small sample. The individual commencement of diminution of the brain is actually highly variable according to statistical law.

Figure 1b shows the cortical surface (Cc-sa) of the human cortex cerebri in relation to brain size with a significant correlation (P = .001). Nevertheless, the variability is considerable.

The values for length and surface of cortical convolutions in man (Fig. 2a) were estimated by Elias et al. (1967b). The correlation of both sizes is significant (P = 0.01). The length of convolutions compared with the age shows only a trend for diminution which is not significant. Between the folding index and the brain size, only a tendency of correlation (P = 0.1) can be found (Fig. 2b). In my opinion, folding is barely influenced by the size of the brain. We will say more about this problem in the Discussion.

Figure 3a compares the surface of cortex (Cc-sa) with the volume of the brain (Br-vo) of some large and small mammals. Each sign in this comparative view is based on one single size of species represented; only in man have we two means of males and females (circled). The interpretation of the results must take into account the individual variability within the order of a species. We may assume that the variation in animals is similar to that in man. The correlation of both values in mammals is exponential and significant with P = .001. The values of man lie slightly below the regression line. This point is worthy of note and will be discussed below.

Figure 3b shows the logarithmic regression between folding index (Cc-fi) and brain size in the mammal. This correlation is highly significant. It is interesting that the humans with two mean values (one for males, one for females) and the elephant are distinctly below the line of regression. This fact will be discussed later on when comparing the volumes of the cortex. The Cc-fi of man is obviously smaller than that of the whales. The dolphins, and especially the *Tursiops truncatus*, with brains of a similar size to man, have Cc-fi which are about 50% higher.

When man and animals are compared as to the size of the cortex cerebri (Cc-vo) relative to age in man and to size of brain of mammals (Br-vo), in humans (Fig. 4a) we find a slight negative but not significant correlation between volume and age. If we divide the values at the age of 60 years, we can calculate the correlations for both groups. We find no correlation (r = .096) at ages up to 65 years. However, after the 60th year of age up to the oldest individual estimated, there is a significantly negative correlation (P = .01) with aging. If we compare this result with the correlation of the total brain, we must consider that the decrease in size of Br-vo is much more obvious than that of Cc-vo. This result is in agreement with other estimates (Haug et al., 1984; Eggers et al., 1984a). Figure 4b contains the comparison of Cc-vo with Br-vo in mammals. The exponential regression is highly significant. In contrast to Figure 3b, the values of man do not deviate from the regression line. This must be seen in connection with the thickness of the cortex cerebri.

In Figure 5, the sizes of the striatum and the thalamus are compared with age in man and with total brain size in mammals. The exponential regression in man (Fig. 5a) shows a highly significant negative correlation for both important nuclei in the subcortical structures. This declining function may be relevant during the total adult life, because we cannot discover a break point of the scatters at the age of 60–65 similar to that for the total brain (Fig. 1a). This result will be discussed later. The correlation of both central nuclei with brain volume in mammals (Fig. 5b) also has a positive significance (P = .001). The human values lie within the regression line.

The Cortical Cytoarchitectonics

The images of the cytoarchitectonics of the areas investigated are very different. The differences are distinctly greater than the individual variability which

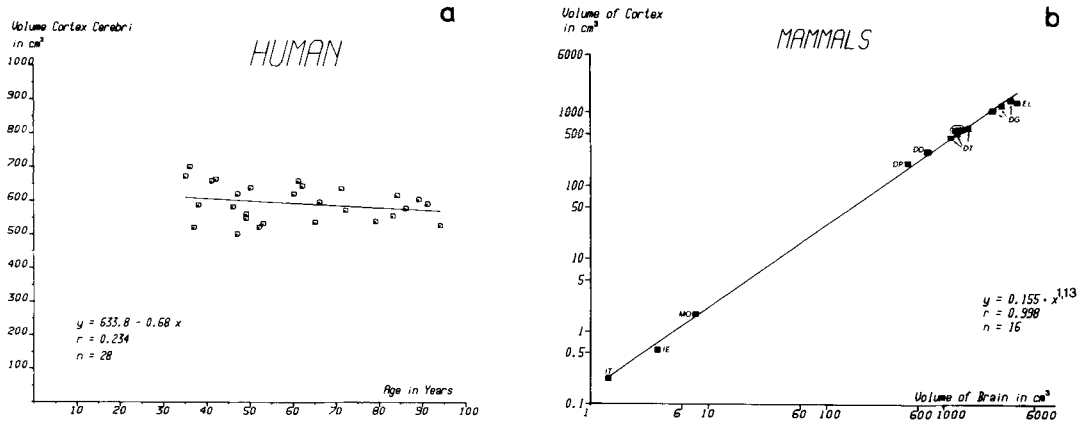


Fig. 4. The volume of cortex in the human is depicted in relation to age and, in mammals, to the volume of the brains (b). **a:** The total correlation in the human is not significant. For ages over 60 years, however, a significance can be found (A = 719, B = −1.68, n = 15, r = .692, P = .01) (values of Haug and coworkers). **b:** There is an exponential regression between both values of high significance (P = .001). The circled values of man lie very near to the regression line (values of Elias and co-workers and Haug and co-workers).

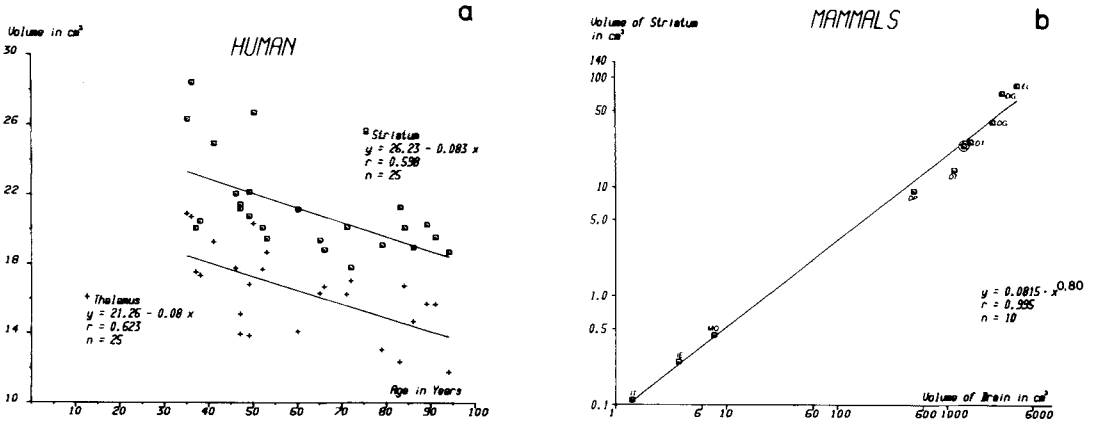


Fig. 5. The volume of the striatum and the thalamus (values of Haug and co-workers). **a:** For humans, volumes are compared with age; the correlation is highly significant for both nuclei (striatum: P = .01; thalamus: P = .001). **b:** The exponential regression of the mammals for the striatum is highly significant (P = .001). The value of man is circled.

was demonstrated by Haug et al. (1984). The cortical areas shown in Figure 6 are not from the same cytoarchitectonic type. But, in my opinion, the cortical enumerations given by Brodmann are actually valid only for the primates. In other mammals, we find areas with the same or similar functions. However, their cytoarchitectonics deviates more or less from the Brodmann's image (1909).

Figure 6 shows that the various mammals differ from each other in some characteristics. The figures regarding the cortex of Odontoceti published by Langworthy (1932), Jansen (1973), Morgane et al. (1980, 1985), Jacobs et al. (1984), and Kesarev et al. (1977) show a great similarity to our own images; this concerns also the individual variability. The following details are important:

1) The thickness of cortex cerebri differs from species to species on the order of two times, e.g., man and Insectivora (hedgehog and mole).

2) Layer I is to some extent very thin, especially in the primates, e.g., *Saimiri*, and partly very thick, e.g., elephant and some dolphins.

3) The density of the neurons is very different. *Microcebus* and *Saimiri* have a very high density, but in the elephant and dolphins it is low.

4) The sizes and size distributions of the neurons are also very variable. We find a high uniformity of the cell sizes, with small cells in *Microcebus* and *Saimiri*; medium cells in the Insectivora, *Tupaia*, and lemurs; and very large cells in the elephant and *Phocaena phocaena*.

The primates including man show a high variability of cell size, but the small granular cells represent a very large fraction.

5) A layer IV which is actually granulated can be seen only in the brains of primates and man. Yet, in the prosimians the granulation is moderately expressed.

Figure 6 can help in understanding the differences in the species with regard to neuronal densities, mean

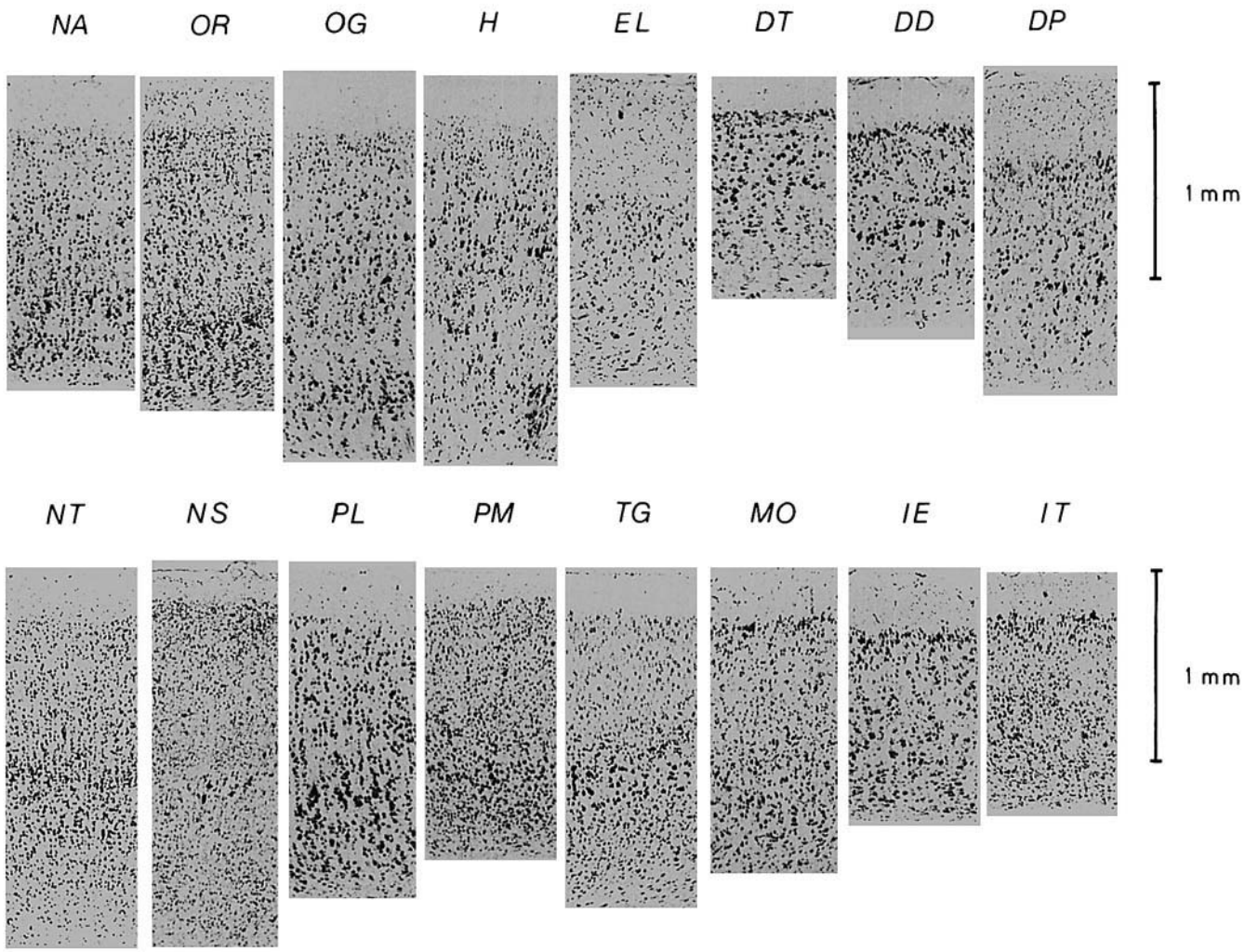


Fig. 6. Cytoarchitectonic images of the cortex cerebri of some mammals and man investigated. All images are taken from the lateral convexity of the frontal lobe concerning probably area 9 (DP, porpoise; H, man; NA, spider monkey; NS, squirrel monkey, NT, night monkey; OG, gorilla; OR rhesus monkey; PL, lemur; PM, mouse lemur, TG, tree shrew.) except in elephant (EL), bottlenose (DT) and common dolphins (DD), opossum (MO), hedgehog (IE), and mole (IT) which were taken from the convexity of the parietal lobe. The comparative areal classification of these animals is not definitely defined.

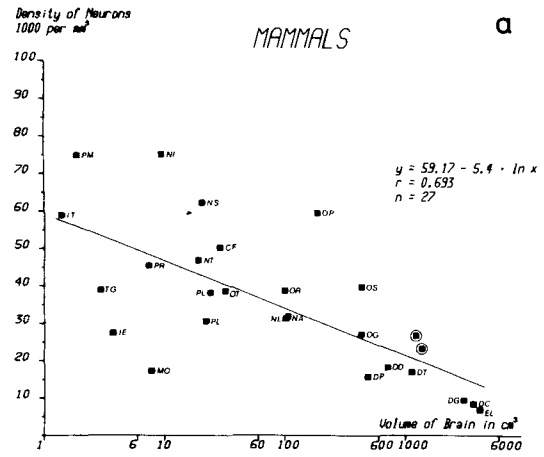
sizes, and size distributions. The graphs show us that cerebral cortices with a high granular fraction are a special evolutionary development, which is—to my knowledge—restricted to those primates beyond the border between prosimians and primates *sensu strictiori.*

The Cortical Neurons: Number and Size

This section deals with the mean values of cells, especially neurons, in the cortex. The evaluations were performed mainly in the convexity of the frontal lobe of the brain. In the dolphins, the elephant, and man, four areas were estimated. A medium arrangement of the investigated areas will be used in comparison. The enumeration will be as follows: density of cells (de), the mean size of neurons (Ne-ms), the volume fraction of the Nissl–stained perikarya (vo%), and the total amount of neurons in the neocortex (Ne-am); the latter was estimated by the volume of cortex and the Ne-de over the cortex, taking the shrinkage into account by multiplication with the residual volume. The size distribution of the cortical perikarya (Ne-sd) will be given separately in the next section.

The cellular densities are calculated for a volume of 1 mm^3. With the procedure of Haug et al. (1971), it is possible to estimate the mean density within the total cortex. Therefore, we measured the cells within rows arranged perpendicularly to the free surface of the cortex. On an average ten to 20 such rows were used for the final results. For each cortex, we measured about 1,000–1,500 neuronal perikarya.

The density of neurons in the neocortex in mammals has a negative correlation with brain volume (Fig. 7a) which is highly significant ($P = .001$). Nevertheless, the variability of the values is distinct. Man and some of the



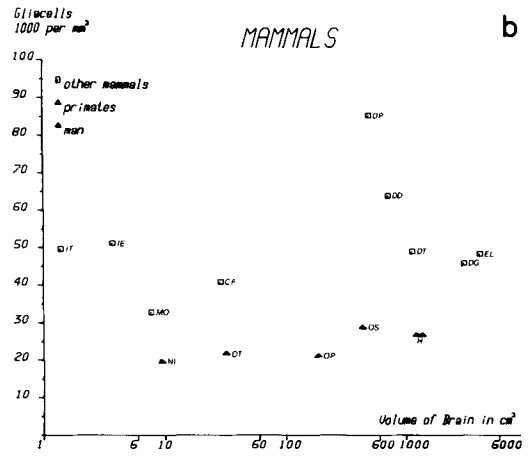
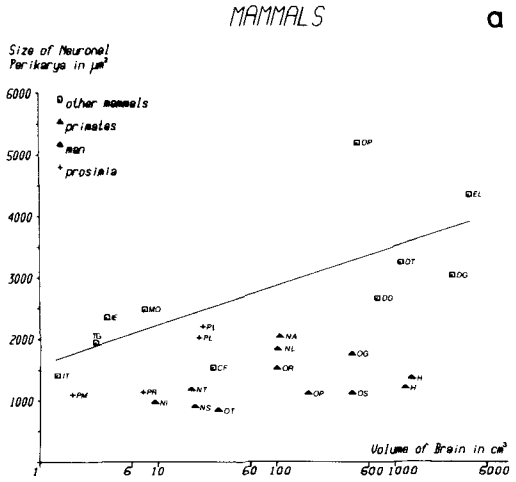


Fig. 7. The cellular density in the cortex cerebri in 1,000 per mm^3 (values of Haug and co-workers). **a:** The neuronal density has logarithmic regression and shows a high significance ($P = .001$) in mammals. The circled human values lie above the regression line. **b:** The density of glial cells in the gray of cortex shows very large variations. A dependence on brain size cannot be observed.



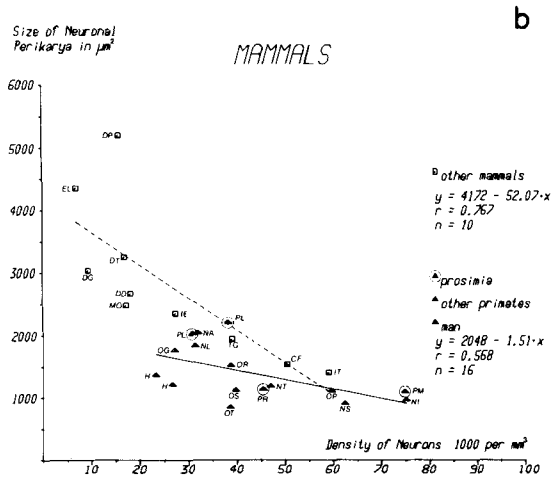


Fig. 8. The size of neuronal perikarya in μm^3 is compared with the size of brain and with the density of neurons (values of Haug and co-workers). **a:** The logarithmic regression shows on the whole no correlation in mammals. The logarithmic regression line does not include primates and man. In primates one cannot see any dependence between the two values. For all other mammals a slight correlation exists (n = 10, A = 1,602, B = 275.8, r = .714, P = .05) **b:** The linear regression of the perikaryon size to the neuronal density is distinctly flatter for primates including prosimians (B = −1.51, $P = .01$) than for the mammals (B = −52.1, $P = .01$).

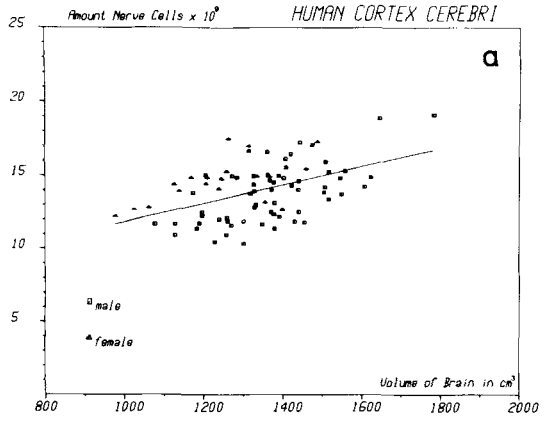
primates lie over the regression line. This characteristic will be commented upon in the Discussion.

The glial density (Gl-de) shows no signs of dependence on the size of the brain in spite of high variance (Fig. 7b). It should be noted that Gl-de is probably lower in primates than in other mammals. Our sample, however, is too small to allow a definitive statement on this question.

Figure 8 compares the mean size of neuronal perikarya (Ne-ms) with the size of brain (Br-vo) in a logarithmic and with Ne-de in an arithmetic regression. The Ne-ms is expressed in μm^3 as volume. The comparison with Br-vo in Figure 8a shows an increasing size; however, the increase is only slightly significant in those mammals ($P = .05$) which are not primates. In primates including man and prosimians, we do not see any relation between Br-vo and Ne-ms.

The correlation of Ne-ms with the Ne-de is negative and significant at the level of .01%. We find differences, however, between primates and the other mammals. The regression in the other mammals is steep and shows a relatively large variance; the one of the primates is relatively flat, yet it has only a small variability. Man lies near the lower left corner, indicating that he has also a small mean size as well as a relatively low density of neurons.

Figure 9 deals with the total amount of neurons (Ne-am) in the individual human neocortex compared with



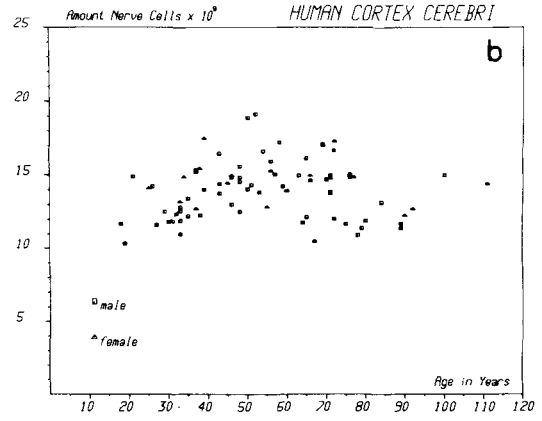
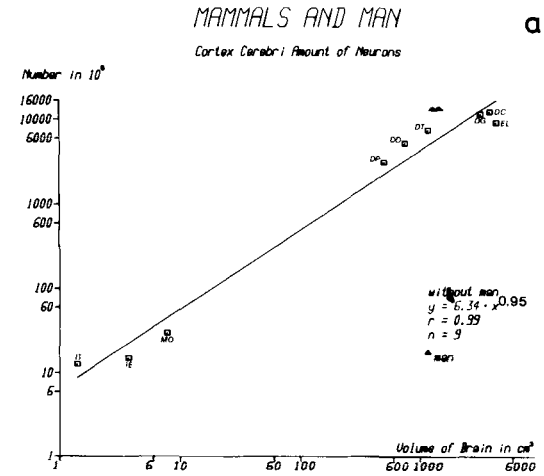


Fig. 9. The individual number of neurons in the human cerebral cortex is compared with the brain size and age. The squares represent males and the triangles females (values of Haug and co-workers). **a:** The brain size shows a linear regression with a significant correlation (n = 81, y = 5.583 + 0.006 x, r = .479, P = .001). **b:** The age is not correlated with the number of neurons.



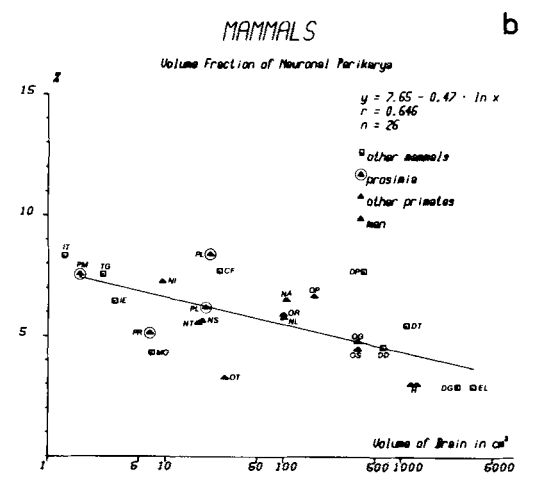


Fig. 10. **a:** The total number of neurons in the cerebral cortices and the brain size of mammals in an exponential regression correlate highly significantly ($P = .001$). **b:** The perikaryon volume fraction and the brain size of mammals in a logarithmic regression correlate highly significantly ($P = .001$).

its brain size and its age. The computation of this value is mentioned in the Methods section. The sample of 81 individual values contains only those specimens for which at least two areas have been evaluated (Haug, 1986).

The Ne-am is very variable (Fig. 9a); however, it increases with brain size, having a great probability of $P = .001$. It is an interesting point that the females, expressed by triangles, lie almost entirely above the regression line. According to their lower brain weight, they lie in the middle and left part of the diagram.

We could find no change of Ne-am during aging nor a difference between male and female (Fig. 9b). Both points were surprising. Our detailed measurement of the effect of aging in four human areas (Haug et al., 1984) has shown that the densities of neurons increase. It is necessary to state that we have eliminated the errors due to the different shrinkage of the human brain tissue with aging and refer to the fresh tissue volume. The values (Ne-am) of the females lying amongst those of the males demonstrate the fact that females have a higher density of neurons in their cortex. Haug (1985) has found that the density is about 4,000 neurons per mm^3 higher in the female than in the male.

Figure 10a shows the exponential regression between the Ne-am in the neocortex and the volume of the brain. The correlation is highly significant at the level of $P = .001$. This fact is to be expected. However, one species, man, deviates clearly above the regression line. Together with the other notable findings, this will be discussed below.

Figure 10b deals with the relation between brain size and the perikaryon-volume fraction (Ne-vo%), which is an expression of the size of the neuropil and, indirectly, of the space in which the dendrites communicate with the axon by synapses. A decrease in Ne-vo% indicates

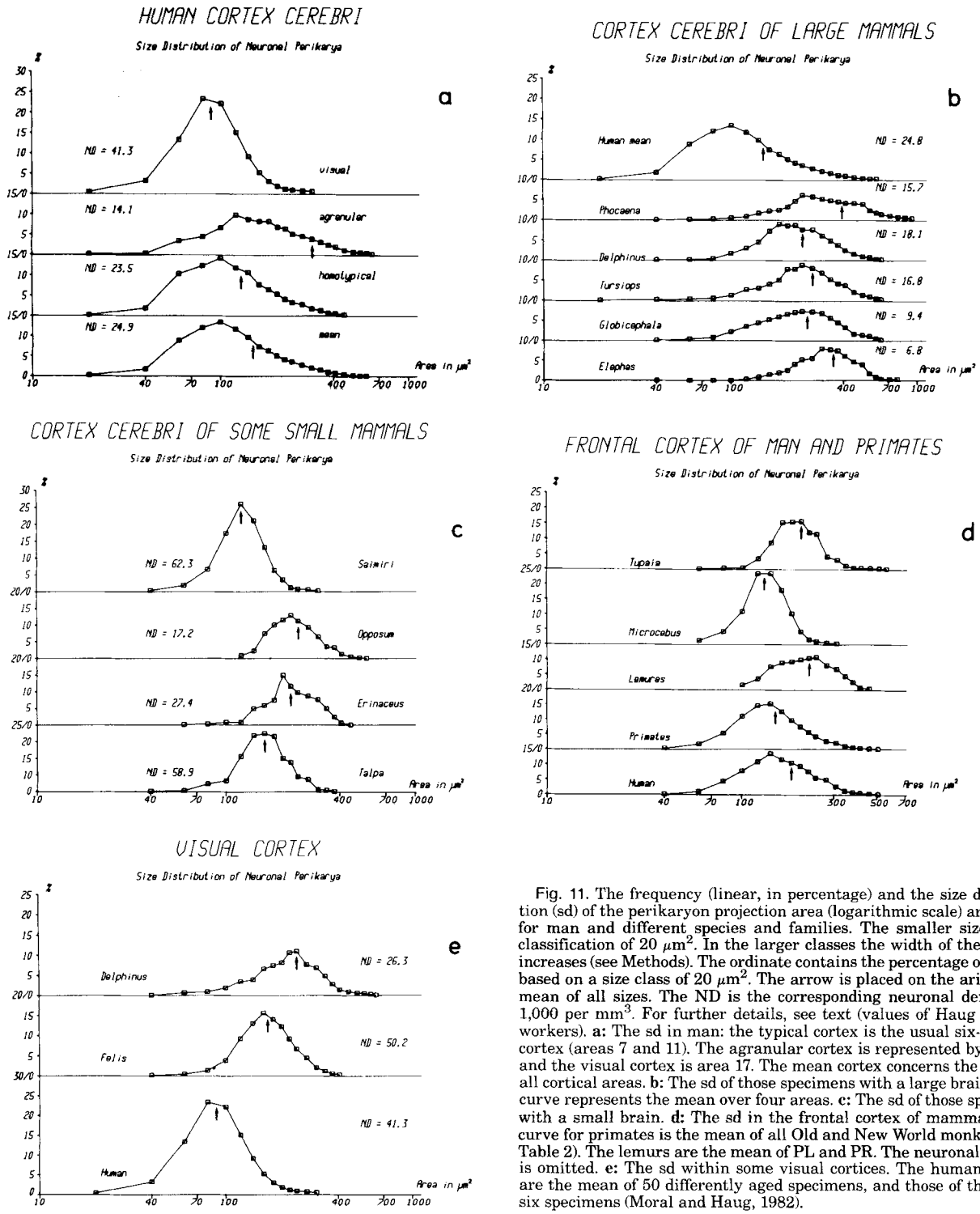


Fig. 11. The frequency (linear, in percentage) and the size distribution (sd) of the perikaryon projection area (logarithmic scale) are given for man and different species and families. The smaller sizes have classification of 20 μm^2. In the larger classes the width of the classes increases (see Methods). The ordinate contains the percentage of values based on a size class of 20 μm^2. The arrow is placed on the arithmetic mean of all sizes. The ND is the corresponding neuronal density in 1,000 per mm^3. For further details, see text (values of Haug and coworkers). **a:** The sd in man: the typical cortex is the usual six-layered cortex (areas 7 and 11). The agranular cortex is represented by area 6 and the visual cortex is area 17. The mean cortex concerns the sd over all cortical areas. **b:** The sd of those specimens with a large brain. Each curve represents the mean over four areas. **c:** The sd of those specimen with a small brain. **d:** The sd in the frontal cortex of mammals. The curve for primates is the mean of all Old and New World monkeys (see Table 2). The lemurs are the mean of PL and PR. The neuronal density is omitted. **e:** The sd within some visual cortices. The human values are the mean of 50 differently aged specimens, and those of the cat of six specimens (Moral and Haug, 1982).

an increase of the neuropil. This measurement manifests a high correlation with brain volume ($P = .001$) in spite of the relatively large variance. A difference in this measurement between the different genera of mammals cannot be found; the primates lie amongst the other mammals.

The results obtained at the cellular level have brought some interesting conclusions which can best be understood if we discuss them collectively.

The Size Distribution of Cortical Neurons

The size distributions of the cortical neurons (Ne-sd) were primarily estimated from a layerlike measurement of flat strips lying parallel to the surface of the cortex. In the study presented, however, such a detailed report on the results cannot be given. Therefore, our results deal with Ne-de in the total depth of the cortex. The curves of Figure 11 demonstrate that important differences exist. The graphs are logarithmic in direction to the size and represent the projection area of the neurons. This logarithmic scale shows that the larger sizes of neurons are apparently represented much less than the smaller ones. The logarithmic scale leads to an increasing width of the size classes, contrary to the visual impression that the class width becomes smaller. The linear heights represent the percentage within the classes; this value is corrected depending on the width of the classes.

The density of the neurons (1,000 per mm^3) is shown on the various parts of Figure 11. This density is highly different for the various species and areas. It should be mentioned, however, that the curves of Ne-sd are based on the percentage and not on the density. Another point is important, i.e., that small Ne-sd have few classes with a high number of neurons. This refers mainly to Ne-sd consisting of small cells. On the other hand, the Ne-sd with large cells spread their values over many classes. Therefore, one can find only a low percentage in each single class.

The arrows in Figure 11 illustrate the mean size of the neurons (Ne-ms) in the cortex. If the Ne-sd is almost normally distributed, the arrow points to the maximum of the Ne-sd. A deviation of the arrow in the direction of the larger size classes indicates a rather logarithmic Ne-sd.

The Ne-sd is based on the measured projection area of the largest circumference of the Nissl-stained neuron. This largest area was found by turning the fine adjustment screw. The cells below the area of 100 μm^2 are smaller than 500 μm^3. Neurons having an area over 200 μm^2 are larger than 1,500 μm^3. At an area of around 400 μm^2, the size increases to over 5,000 μm^3; and at 600 μm^2, a size of over 10,000 μm^3 is reached. The largest neurons measured in the *Phocaena phocaena* (DP) were in the range of 950 μm^2 with a volume of over 20,000 μm^3. The granular cells are below 400 μm^3; this equals a projection area of 90 μm^2. The smaller pyramidal neurons are between 400 and about 1,000 μm^3. The largest cells in DP have the size of the smaller giant Betz neurons of man.

The average Ne-sd within the human cortex (Fig. 11a) was estimated in four different areas according to the main types of cytoarchitectonics. In man many of the smaller neurons are of the granular type, and most of the large neurons are of the pyramidal type. Each area indicates the average of at least 50 specimens. The curves represent very different types of Ne-sd. The human visual cortex (area 17) presents a high percentage of neurons within the smaller size classes and very few cells in the larger classes. The total Ne-de is in the range of 41.3 (1,000 neurons per mm^3). The agranular cortex of the extrapyramidal area 6 has an average Ne-de of only 14.1; this is about a third of the density of area 17. It shows fewer neurons only in the smaller classes. The larger classes are represented by a relatively long tail in the direction of the higher size classes. The typical six-layered cortex (areas 7 and 11) and the mean cortical representation of Ne-sd in the total cortex are very similar.

The impression from these Ne-sd curves of the human cortex is of the variability within one species. This is important for the comparison with other genera of mammals.

In Figure 11b we can see Ne-sd of large mammals like dolphins and the elephant. They have only a few small neurons. The distributions expand distinctly in the large and very large size classes. The Ne-sd within the dolphins *sensu strictiori (Delphinus delphinus, Tursiops truncalus,* and *Globicephala macrorryncha)* are similar. On the other hand, the small *Phocaena*, with only a 500-cm^3 brain size, shows a Ne-sd consisting of many very large cells. They are distinctly larger than those of the elephant, which, however, has the second-largest Ne-sd of all mammals investigated.

The comparison with man shows that the mammals with large brains have mainly large cells of the pyramidal type in their cortices, while in the size classes below the projection area of 100 μm^2, man has a lot of small cells which are probably mostly of granular types. The density of neurons in man is distinctly higher than in mammals with large brains. The elephant has, with 6.8 ($\times$ 1,000/mm^3), the lowest Ne-de measured in our investigation. The consequences of this kind of differentiation of composition will be commented on below.

In animals with smaller brains, the Ne-sd are relatively small (Fig. 11c). This means that neuronal sizes are very uniform. However, small mammals, except the *Saimiri* (primate, platyrrhini), do not have really small or very small neurons. It seems that they have no or only a few granular cells. The location of the arrow for the mean size in all these mammals lies near the maximum of the Ne-sd, which speaks for a normal distribution. The *Saimiri* will also be mentioned below.

Figure 11d deals with the primates. All results concern the convexity of the frontal cortex and in man, area 11. In this graph the arrows for the Ne-de were omitted, because the curves for the lemurs and primates are combinations of more than one species. The Ne-sd found in all Old and New World monkeys were very similar. Therefore, I have used a common curve which includes five Old World and five New World monkeys; the species are listed in Table 2. The common curve for the primates shows that the neurons smaller than 100 μm^2 constitute about 20%. This is similar to the human frontal area 11, which, on the other hand, also has a remarkable number of larger cells (location of the arrow). The Ne-sd of the very small prosimian *Microcebus* consists of many cells around the projection area of 120 μm^2. This supports the

premise that none or only a few granular cells are joined. The Ne-sd in the lemurs (*Lemur catta* and *Lemur ruficaudatus*) is nearly the same. Both large prosimians have no granule cells, and their cortex consists mainly of medium-large pyramidal cells.

Tupaia also has, similar to the lemurs, only medium to large cells. The graph of Figure 11d shows that the Ne-sd of the primates (Old and New World monkeys) and man are greatly similar. Both zoological families have a cortical cytoarchitectonic structure which is characterized by a large fraction of small, mainly granular cells.

The influence of the fraction of granular cells on Ne-sd is demonstated in Figure 11e, which compares three visual cortices. In man, over 50% of all cells are smaller than 100 μm^2. Economo (1926) has therefore introduced the name "coniocortex." In the cat we find a neuronal density which is similar to that of man; but Ne-sd is quite different, because the number of small cells is much lower. The visual cortex is constructed mainly of medium-to-large cells, similar to the pyramidal type. In the common dolphin, the size distribution is greater than in man and the cat. The cortex is small and consists of pyramidal cells. A similar construction can be found in other dolphins (*Tursiops* and *Globicephala*). Their visual cortices show a decreasing density and an increasing neuronal size with the enlargement of the brain.

The figures for the Ne-de in the cortex cerebri of some mammals and man look similar. However, the composition as to type of neurons must be very different in the various families; this is true especially for primates and man.

DISCUSSION

As to the general meaning of our results, attention should be paid to the fact that the values obtained are sampled from mammals of very different families. The different procedures used in various measurements by other authors must be considered and a basis found for the interpretation of data. I shall divide the account into three parts. First are the special behavioral patterns within one species regarding variation and biological changes in aging. In this way we receive information about the various measurements and their changes which may influence the results obtained from one individual of a single species. It must be stated that our results are corrected for embedding shrinkage and its alteration during aging; therefore, they represent values for fresh material.

In the second part, the common rules within the mammals studied will be discussed. This part must be connected with the usual assumption of the size of structure parameters by the so-called allometric method (Jerison, 1974). Combined with this view is the third part, which deals with the special developments in primates and man. In this connection, the comparison of those animals with large brains and high animal intelligence is of great interest.

Our investigation shows that the variation in all size measurements is great. The variance of the interindividual macroscopic values expressed in percentage of standard deviation in man is 10–15%. Within this range lie about two-thirds of all values of one species. The size variation in the cellular structures is between 15 and 25%. The methodical error is usually smaller than 5% during our estimation concerning larger structures or structural components.

If we transfer this variation to the single species of mammals in our investigation we are able to say that differences in the body and brain size between the individuals of one species and the average of the species are certainly larger than 20%. The differences of the measurements between the various species of mammals are larger than 1,000 times. For example, *Sorex minutus* has a brain weight of 0.25 gm and *Physeter catodon*, 9,000 gm (Haug, 1970a). Therefore the comparison of our results is less influenced by the individual variation and the methodical problems. Thus it is reasonable to draw comparative conclusions in quantitative terms.

A further point is important to mention: the age-dependence changes in humans are very complex. The size of the brain diminishes after the age of 60–65 years (Eggers et al., 1984a, Haug, 1984); however, a large individual variance can be observed with respect to the age of onset. During biological (physiological) aging, we cannot observe a diminution in the number of neurons in the human cerebral cortex. The shrinkage of the brain during aging, however, leads to an increasing density of neurons. In other words, the neurons move closer together during aging. This observation transfers the changes from the rough cellular to the subcellular level during aging (Haug, 1984, 1985; Haug et al., 1984).

A further point must be noted. The age changes in the human brain are very different. Those parts of the cerebral cortex which are closely connected with the sensory functions show fewer signs of aging (area 7 and 17), while the motor function areas are strongly affected. This can be seen in the extrapyramidal area 6 in which the perikarya shrinks to about two-thirds (Eggers et al., 1984a). An interesting observation in humans is the fact that females and males have the same total number of neurons, although the size of the female brain is about 100–150 cm^3 smaller than that of the male. The neuronal densities in all five areas investigated were statistically significantly higher in females (Haug, 1985). It is unknown whether similar differences between the sexes can be found in all higher mammals. This question is discussed by Swaab and Hofman (1984) and Döhler and Gorski (1981).

The comparative results at the macroscopic level show, for all parameters estimated, a clear relation to brain size. This refers to the surface of cortex (Cc-sa) and the volume of cortex (Cc-vo), thalamus, and striatum. All estimated correlation functions are significant at the level of $P = .01$ or .001.

The previous investigations of Bok (1959), Stephan (1972), Stephan and Andy (1969), and Bauchot (1978) have demonstrated a similar behavior. However, the examined primates, rodents, and insectivores represent only a small sector of the mammals. These scientists did not have the opportunity to measure mammals with very large brains. Furthermore, they compare the animals mainly by using brain and body sizes and selected indicators such as "progression indices" for brain and different brain parts like neocortex. In our results (Figs. 3a, 4b, 5b), the exponential regression has a slope of

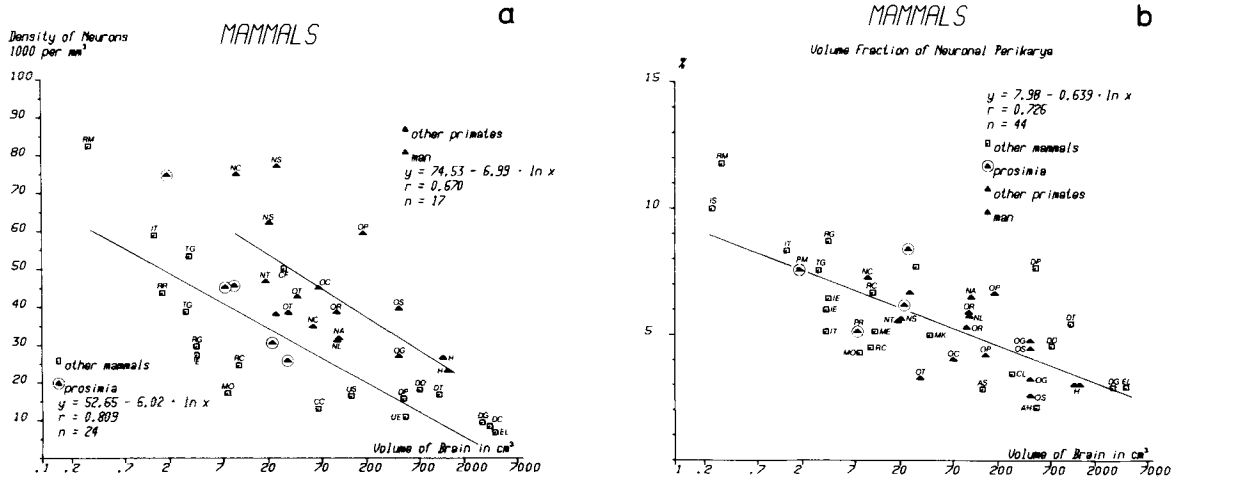


Fig. 12. The size of the brain and its logarithmic correlation to neuronal density and the neuronal volume fraction; the data are enriched with older results and values of Ryzen and Campbell (1955) and Tower (1954). **a:** Density of neurons. Primates and man have a significance of $P = .01$ and other mammals including the prosimians of $P = .001$. **b:** The perikaryon-volume fraction is at $P = .001$ and is highly significant for all mammals.

about 1.1 except in the striatum. This may be proof that in large brains the Cc increases more strongly than the basal ganglia including striatum.

One realizes, however, that the values of the humans are distinctly smaller regarding the surface area and folding index of the cerebral cortex than expected according to the regression function for all mammals including the large Odontoceti (Elias and Schwartz, 1969, 1971; Elias et al., 1967a,b, 1969; Haug, 1970a). On the other hand, humans lie within or slightly over the regression line for the volume of cortex. This accords with the results of Haug (1970a), who has found out that Cc-vo has, in man, the largest size of about 47%, while the Cc-vo of the Odontoceti is only 43–44%.

Consequently, the thickness of the human Cc should be greater than that of mammals with large brains, including the Odontoceti and the elephant. The relatively low Cc-vo of the elephant is due mainly to its very large cerebellum. It has, at about 30%, the largest volume fraction of the cerebellum of all mammals (Haug, 1970a). According to our own measurement, the thickness of Cc (see Fig. 6) in man is distinctly greater than in other mammals. It is striking that the layer I of the large mammals is very thick. Figure 6 shows that this layer amounts to over one-third of the total Cc in the elephant. The report on the thickness of Cc in Odontoceti, especially in *Tursiops*, also shows a thickness between 1.2 and 1.6 mm (Langworthy, 1932; Jacobs et al., 1984; Jansen, 1973; Kesarev et al., 1977; Morgane et al., 1980, 1985). This result is not in accordance with Prothero and Sundsten (1984), who report on an increasing cortical thickness with brain size.

The macroscopic comparison shows that many authors have drawn conclusions as to the special situation of man without enough knowledge of large brains, e.g., whales (Bok, 1959; Stephan, 1972; Stephan et al., 1984, Hofman, 1984). With regard to surface and folding, man is less developed only compared with dolphins.

The cellular level, however, gives a quite different aspect. The glial cell density (Gl-de) probably does not depend on the brain size. A difference between males and females does not exist. The Gl-de in the gray part of the cortex cerebri is at the range of 30,000–60,000 per mm^3. The Gl-de in primates is possibly lower than in other mammals, yet we cannot find a statistical significance.

In spite of a high variance, the density of the neurons (Ne-de) has a clear connection with brain size, where density decreases while brain size increases. However, many of the primates and man lie above the regression line. In addition to the values presented in Figure 7a, Figure 12a contains older estimations from our own and other groups. The separately calculated regression functions for primates, including man, and the other mammals show that both groups have a similar slope (−6.0 and −7.0). The regression for the primates, however, is about twice as high as that for the other mammals. Similar results were reported upon by Tower (1954) and Haug (1967b, 1981b).

The mean size of the neuronal perikarya (Ne-ms) has no simple relation to brain size. The Old and New World monkeys (Figs. 6, 8b, 11d) have similar Ne-ms and size distributions of neurons (Ne-sd). In the other mammals, however, Ne-ms increases with brain size (Bok, 1959). Another special evolution in monkeys and man is the high fraction of cortical granular neurons with switching functions. Only in primates does layer IV have a typical granular formation consisting mainly of such small cells. All other mammals have a layer IV with neurons of medium size. Nevertheless, the fibers lying parallel to the surface in layer IV represent a common development of nearly all mammals.

One interesting finding is common to all mammals: this is the size of perikaryon-volume fraction (Ne-vo%), or the reciprocal gray-cell coefficient which represents the size of the neuropil in Cc. The Ne-vo% decreases

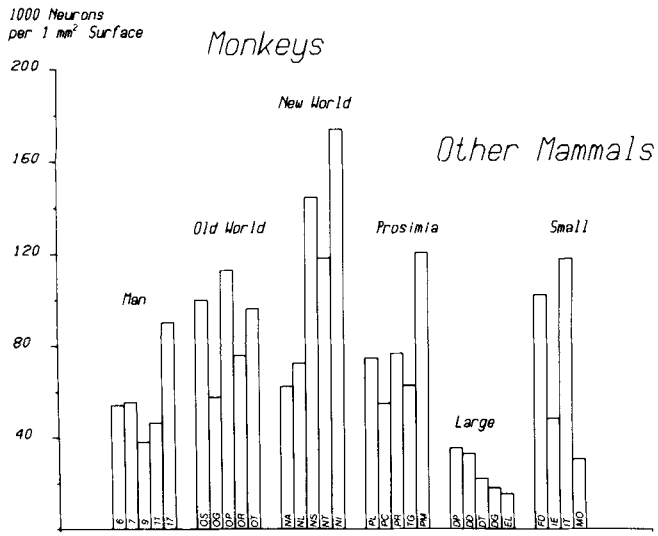


Fig. 13. The number of neurons in the cortex beneath a cortical surface area of 1 mm^2. For details see text.

with increasing brain size (Haug, 1967a, 1981b). The significance of this decrease is the corresponding increase of the neuropil, which means that the number of interconnections between the perikarya also increases. This corresponds to the larger number of neurons in the total brain. Figure 12b contains data from some older investigations besides our own values. The variation of values is relatively high; it is a question of whether this variation contains differences between the mammalian families. If such differences of Ne-vo% exist, they will be relatively small.

The material of our investigation permits an answer to the question of whether the number of neurons in a certain cylinder of Cc is independent of species and family. Bok and Taalman Kip (1939) and Rockel et al. (1980) have postulated that this number is, in all mammalian Cc, the same below a certain surface area of Cc. Only the visual cortex has about double the number in such a cylinder. Our square is, with a 1-mm edge size, about 1,111 times larger than that calculated by Rockel et al. (1980). On the other hand, we also considered the three-dimensional shrinkage.

Our own results concerning Ne-de (1,000 per mm^2) can almost be compared with those of Bok and Taalman Kip by multiplication by a factor of 60 and with Rockel's by multiplication by 1,900. Both factors reflect the mean shrinkage and the different sizes of surface area of Cc. Figure 13 shows our results. The range extends from about 19 (1,000 neurons per 100 μm^2 times cortex thickness) in the elephant up to 190 in the woolly monkey (NL). Also in man we have found large differences in neuronal density, between 37 and 92. An unpublished personal recalculation of the values of Economo and Koskinas (1925) for man has found Ne-de of 35 in area 8 and over 100 in all primary sensory Cc, with a total mean of 50. I think that Rockel's assumption is restricted only to primates, some rodents, and carnivores. It cannot be a general rule for all mammals.

In conclusion, we may state, there is a certain framework of allometric connection within the brain and Cc structures within the order of mammals. The allometric correlation is relatively high concerning the macroscopic measurements. However, the different families deviate from this main correlation in a special direction for each family (Jerison, 1974). The primates have evolved a deviation in the direction of a high fraction of granular neurons in Cc. Furthermore, their cortices are distinctly thicker compared with other mammals. The result is an increasing complexity in the vertical columns of the cortex. The Odontoceti, however, have not developed a granular neuronal system. They have increased the Cc folding distinctly more than the primates; however, we do not know anything about their cortical columns.

The high folding of Cc and the relatively thin Cc in whales may be a development which has, on the whole, the same functional significance as the development in primates. Furthermore, it is possible that the low thickness of Cc in the whales has a greater window function of convolution (Prothero and Sundsten, 1984). The window function is connected with the number of fibers which can pass into the space between the gray matter of the convolution. This assumption explains the development of large brains in whales. Prothero and Sundsten (1984) postulated for a 2-mm-thick Cc a brain size of about 2,800 cm^3 for an optimal function. But according to our own observation and those of Kesarev et al. (1977) and Morgane and Jacobs (1972), the whales have a Cc thickness distinctly less than 2 mm.

Hans Elias with his remarkable engagement in stereological procedures in general, and his introduction of macroscopic measurements of brain structures as to surface and length of the cerebral convolution, helped to obtain a new insight into brain composition. I, personally, have been greatly impressed by his work, and our association influenced my macroscopic and microscopic survey of the cortex of man with respect to age, variation, and comparison to other mammals.

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LITERATURE CITED

Barmwater, U. 1982 Stereologische Untersuchungen über die Alterung der extrapyramidalen Hirnrinde des Menschen (Brodmann Area 06), verbunden mit einer Analyse über die Genauigkeit und Konstanz der Dicke von histologischen Schnitten. Dissertation, Med. University Lübeck.

Bauchot, R. 1978 Encephalization in vertebrates. A new mode of calculation for allometry coefficients and isoponderal indices. Brain Behav. Evol., *15*:1–18.

Blinkov, S.M., and I.I. Glezer 1968 The human brain in figures and tables. Basic Books, New York.

Bok, S.T. 1959 Histonomy of the Cerebral Cortex. Elsevier Publishing, Amsterdam.

Bok, S.T., and M.J. van Erp Taalman Kip 1939 The size of the body and the size and the number of nerve cells in the cerebral cortex. Acta Neerl. Morphol., *3*:1–22.

Brodmann, K. 1909 Vergleichende Lokalisationslehre der Großhirnrinde. J.A. Barth, Leipzig.
Decker, H. 1979 Ein Beitrag zur Methodik der makrosopisch-stereologischen Bestimmung von Volumenanteilen und Oberflächen am menschlichen Gehirn. Disseration, Med. University Lübeck.
Döhler, K.-D., and R.A. Gorski 1981 Sexual differentiation of the brain: Past, present and future. BRI Bull., *5*:5–9.
Dubois, E. 1914 Die gesetzmäßigen Beziehungen von Gehirnmasse zur Körpergröße. Z. Morphol. Anthropol., *18*:323–350.
Economo, C. v. 1926 Ein Koeffizient für die Organisationshöhe der Grosshirnrinde. (Zellanzahl derselben und einige andere Cortex-Maße.). Klin. Wochenschr., *5*:593–595.
Economo, C. v., and G.N. Koskinas 1925 Die Cytoarchitektonik der Hirnrinde des erwachsenen Menschen. Springer, Wien, Berlin.
Eggers, R., H. Haug, and D. Fischer 1984a Preliminary report on macroscopic age changes in the human prosencephalon. A stereologic investigation. J. Hirnforsch., *25*:129–139.
Eggers, R., H. Haug, and Ch. Orün 1984 Präparative Einflüsse auf das Schrumpfungsverhalten und das mikroskopische Bild von Gehirngewebe. Verh. Anat. Ges., *78*:529–531.
Elias, H. 1962 Stereology, scope and limits. Stereology, *1*:1–2.
Elias, H. 1967 Introduction: problems of stereology. In: Stereology: Proceedings of Second International Congress for Stereology. H. Elias, ed. Springer, New York, pp. 1–11.
Elias, H., and D. Schwartz 1969 Surface areas of the cerebral cortex of mammals determined by stereological methods. Science, *166*:111–113.
Elias, H., and D. Schwartz 1971 Cerebro-cortical surface areas, volumes, length of gyri and their interdependence in mammals, including man. Z. Säugetierkunde, *36*:147–163.
Elias, H., H. Haug, W. Lange, and D. Schwartz 1967a Cerebro-cortical surface areas of some mammals. Am. Zoologist, *7*:291.
Elias, H., St. Kolodny, and D. Schwartz 1967b Surface area and length of convolutions of the cerebral cortex. In: Stereology: Proceedings of the Second International Congress for Stereology. H. Elias, ed. Springer, New York, pp. 77–78.
Elias, H., H. Haug, W. Lange, G. Schlenska, and D. Schwartz 1969 Oberfächenmessungen der Großhirnrinde von Säugern mit besonderer Berücksichtigung des Menschen, der Cetacea, des Elefanten und der Marsupalia. Anat. Anz., *124*:461–463.
Fischer, D. 1982 Altersabhängige Volumen- und Oberflächenveränderungen definierter corticaler und nichtcorticaler Strukturen des menschlichen Grosshirns—Eine makroskopische Untersuchung mit stereologischen Methoden. Dissertation, Med. University Lübeck.
Floderus, S. 1944 Untersuchungen über den Bau der menschlichen Hypophyse mit besonderer Berücksichtigung der quantitativen mikromorphologischen Verhältnisse. Acta Pathol. Microbiol. Scand. [Suppl.], 53.
Gundersen, H.J.G. 1978 Estimators of the number of objects per area unbiased by edge effects. Microsc. Acta, *81*:107–117.
Gunschmann, R. 1972 Vergleichende Studie über die Nervenzelldichte der Großhirnrinde von Mensch, Elefant und einigen Delphinoideae. Dissertation, Christian-Albrechts-Universität Kiel.
Haug, H. 1953 Der Grauzellkoeffizient des Stirnhirnes der Mammalia in einer phylogenetischen Betrachtung. Acta Anat. (Basel), *19*:60–100, 153–190, 239–270.
Haug, H. 1956 Remarks on the determination and significance of the gray cell coefficient. J. Comp. Neurol., *104*:473–492.
Haug, H. 1958 Quantitative Untersuchungen an der Sehrinde. G. Thieme, Stuttgart.
Haug, H. 1960 Die quantitativen Zellvolumenverhältnisse der Hirnrinde. Ein Vergleich der menschichen Ontogenese mit den Verhältnissen bei den Mammalia. In: Structure and Function of the Cerebral Cortex. D.B. Tower and J.P. Schade, eds. Elsevier, Amsterdam, pp. 28–35.
Haug, H. 1967a Zytoarchitektonische Untersuchungen an der Hirnrinde des Elefanten. Anat. Anz., *120*:331–337.
Haug, H. 1967b Über die exakte Feststellung der Anzahl Nervenzellen pro Volumeneinheit des Cortex cerbri, zugleich ein Beispiel für die Durchführung genauer Zählungen. Acta Anat. (Basel), *67*:53–73.
Haug, H. 1970a Der makroskopische Aufbau daes Großhirns. Qualitative und quantitative Untersuchungen an den Gehirnen des Menschen, der Delphinoideae und des Elefanten. Ergebn. Anat. Entwicklungsgesch., 43/4.
Haug, H. 1970b Quantitative data in neuroanatomy. Prog. Brain Res., *33*:113–127.
Haug, H. 1976 Die Zelldichte im Cortex der Primaten und ihre Bedeutung für die Phylogenese der Schaltzellen. Verh. Anat. Ges., *70*:253–258.
Haug, H. 1979 The evaluation of cell-densities and of nerve-cell-size distribution by stereological procedures in a layered tissue (cortex cerebri). Microsc. Acta, *82*:147–161.
Haug, H. 1981a On the proper use of point-counting and semi-automatic procedures in stereology. Microsc. Acta, *85*:141–152.
Haug, H. 1981b Die Evolution der Interneurone in der Hirnrinde von Affen und Delphinen sowie ihre Bedeutung für komplexe Funktionen einschließich der Sprache. Nova Acta Leopoldina N.F., *245*:763–777.
Haug, H. 1982 The location and size distribution of neurons in the layered cortex. Acta Stereol., *1*:259–267.
Haug, H. 1984 Der Einfluß der säkularen Akzeleration auf das Hirngewicht des Menschen und dessen Änderung während der Alterung. Gegenbaurs Morphol. Jahrb., *130*:481–500.
Haug, H. 1985 Are neurons of the human cerebral cortex really lost during aging? A morphometric examination. In: Senile Dementia of the Alzheimer Type. J. Traber and W.H. Gipsen, eds. Springer, Berlin, pp. 150–163.
Haug, H. 1986 History of neuromorphometry. J. Neurosci. Methods *18*:1–17.
Haug, H., J. Kebbel, and G.-L. Wiedemeyer 1971 Die Messung der mittleren Zelldichte und ihre Verteilung in Geweben mit erheblichen Zelldichteunterschieden. Auswertung am Cortex cerebri als Beispiel. Microsc. Acta, *71*:121–128.
Haug, H., S. Kühl, E. Mecke, N.-L. Sass, and K. Wasner 1984 The significance of morphometric procedures in the investigation of age changes in cytoarchitectonic structures of human brain. J. Hirnforsch., *25*:353–374.
Hennig, A. 1958 Kritische Betrachtungen zur Volumen- und Oberflächenmessung in der Mikroskopie. Zeiss-Werkzeitschrift, *30*:3–12.
Hofman, M.A. 1984 Towards a General Theory of Encephalization. An Allometric Study on the Evolution and Morphogenesis of the Mammalian Brain. Academisch Proefschrift, Amsterdam.
Jacobs, M.S., A.M. Galaburda, W.L. McFarland, and P.J. Morgane 1984 The insular formations of the dolphin brain: Quantitative cytoarchitectonic studies of the insular component of the limbic lobe. J. Comp. Neurol., *225*:396–432.
Jansen, J. 1973 The central nervous system in Cetacea. Nautilus Ciba-Geigy:7–8.
Jerison, H.J. 1974 On the meaning of brain size. Science, *184*:677–679.
Kesarev, V.S., L.I. Malofeyeva, and O.V. Trykova 1977 Ecological specificity of cetacean neocortex. J. Hirnforsch., *18*:447–460.
Kühl, S. 1981 Altersabhängige Veränderungen im Gyrus rectus des Menschen. Eine quantitative cytoarchitektonische Untersuchung der Area 11 von zwanzig Gehirnen Männlicher Individuen, verbunden mit einer Analyse von Nervenzellformen zur morphometrischen Bestimmung von Perikaryon-volumina. Dissertation, Med. University Lübeck.
Kühl, S., H. Haug, and W. Schließer 1982 Morphometry of cortical neurons. The best estimation of perikaryon volume from the projection area. Microsc. Acta, *86*:315–322.
Langworthy, O.R. 1932 A description of the central nervous system of the porpoise (*Tursiops truncatus*). J. Comp. Neurol., *54*:437–500.
Moral, J.R., and H. Haug 1982 The monocular and binocular segment of cat's visual cortex. A cytoarchitectonic study with stereological procedures. Anat. Embryol. (Berl.), *163*:487–500.
Morgane, P.J., and M.S. Jacobs 1972 Comparative anatomy of the Cetacean nervous system. In: Functional Anatomy of Marine Mammals. R.J. Harrison, ed. Academic Press, London, pp. 117–244.
Morgane, P.J., M.S. Jacobs, and W.L. McFarland 1980 The anatomy of the brain of the bottlenose dolphin (*Tursiops truncatus*). Surface configurations of the telencephalon of the bottlenose dolphin with comparative anatomical observations in four other Cetacean species. Brain Res. Bull., *5*:Suppl. 3.
Morgane, P.J., M.S. Jacobs, and A. Galaburda 1985 Conservative features of neocortical evolution in dolphin brain. Brain Behav. Evol., *26*:176–184.
Nass, E. 1985 Altersabhängige Veränderungen im Lobus temporalis (Area 20 nach Brodmann) des Menschen. Eine lichtmikroskopische Untersuchung mit morphometrischer Bestimmung von Zelldichte und Zellgröße, verbunden mit einer Betrachtung von Geschlechtsunterschieden. Dissertation, Med. University Lübeck.
Pirlot, P. 1974 Size and activity of brain-structures. Z. Zool. Systematik Evolutionsforsch., *12*:152–155.
Prothero, J.W., and J.W. Sundsten 1984 Folding of the cerebral cortex in mammals. A scaling model. Brain Behav. Evol., *24*:152–167.
Rockel, A.J., R.W. Hiorns, and T.P.S. Powell 1980 The basic uniformity in structure of the neocortex. Brain, *103*:221–244.

Ryzen, M., and B. Campbell 1955 Organization of the cerebral cortex. III. The cortex of *Sorex pacificus*. J. Comp. Neurol., *102*:365–423.

Sass, N.-L. 1982a Altersabhängige Veränderungen im Gyrus rectus des Menschen nach dem 35. Lebensjahr. Eine lichtmikroskopisch-morphometrische Untersuchung zur Änderung von Zelldichten und Perikaryon-volumina, verbunden mit einer Analyse des Verhaltens der Einbettungs-schrumpfung in verschiedenen Lebensaltern und weiteren Überlegungen zur Methodik. Dissertation, Med. University Lübeck.

Sass, N.-L. 1982b The age-dependent variation of the embedding-shrinkage of neurohistological sections. Mikroskopie (Wien), *39*:278–281.

Schlenska, G. 1969 Messungen der Oberfläche and der Volumenanteile des Gehirnes menschlicher Erwachsener mit neuen Methoden. Z. Anat. Entwicklungsgesch., *128*:47–59.

Stephan, H. 1960 Methodische Studien über den quantitativen Vergleich architektonischer Struktureinheiten des Gehirns. Z. Wiss. Zool., *164*:143–172.

Stephan, H. 1972 Evolution of Primate brains: A comparative anatomical investigation. In: The Functional and Evolutionary Biology of Primates. R. Tuttle, ed. Aldine/Ainerton Inc., Chicago, Vol. 6, pp. 155–174.

Stephan, H. 1976 Vergleichende Anatomie des Allocortex. Verh. Anat. Ges., *70*:217–251.

Stephan, H., and O.J. Andy 1969 Quantitative comparative neuroanatomy of primates: An attempt at a phylogenetic interpretation. Ann N.Y. Acad. Sci., *167*:370–387.

Stephan, H., H.D. Frahm, and G. Baron 1984 Comparison of brain structure volumes in Insectivora and primates IV. Non-cortical visual structures. J. Hirnforsch., *25*:385–403.

Swaab, D.F., and M.A. Hofman 1984 Sexual differentiation of the human brain. A historical perspective. Prog. Brain Res., *61*:361–374.

Tjan, Th.-F. 1974 Vergleichende Zytoarchitektonik des Cortex cerebri von Zahnwalen und Elefant. Dissertation, Christian-Albrechts-Universität Kiel.

Tower, D.B. 1954 Structural and functional organization of mammalian cerebral cortex: The correlation of neuron density with brain size. J. Comp. Neurol., *101*:19–51.

Wagner, R. 1864 Messungen der Hirnoberfläche mit besonderer Beziehung auf Windungs- und Furchenbildung und deren Beziehung zur Intelligenz. Abh. Ges. Wiss. Göttingen, *10*:90–97.

Weibel, E.R. 1979 Stereological Methods. Practical Methods for Biological Morphometry. Academic Press, London.

Wiedemeyer, G.-L. 1974 Vergleichende stereologische Bestimmung der Massen und der Oberflächen der verschiedenen Großhirnanteile sowie der Zelldichte bei niedrigen Säugern (Opossum, Igel, Maulwurf). Z. Mikrosk. Anat. Forsch, *88*:299–324.

Wirz, K. 1950 Studien über die Cerebralisation: Zur quantitativen Bestimmung der Rangordnung bei Säugetieren. Acta Anat., *9*:134–196.